COURSE SLIDES

Build a Multi AI Agents Workflow for Stock Trading (C177)

Course Code: C177

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE
Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION
PhD; ACTA; DACE; Executive Programme in Algorithmic Trading (QuantInsti)

DELIVERS
agentic AI, multi-agent systems, AI vibe coding, algorithmic trading and quantitative finance

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

1. Prerequisite check: what did you run in AI Powered Quantitative Trading Fundamentals (C728)?
2. Your name, your role and what you trade or analyse (stocks, ETFs, none yet).
3. Which AI coding assistant have you used - Claude Code, Codex, Cursor, Copilot or none?
4. One trading decision you would like an AI agent to help you make.

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HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. Learn by doing — try every hands-on lab.

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COURSE PORTAL

Download Course Material

1. Sign in at the LMS-TMS portal
2. Open this course from your dashboard
3. Go to the Courseware tab
4. Download the Slides (PPT/PDF), Learner Guide and Lesson Plan
5. Keep them open alongside the labs as you work

Portal: https://lms-tms.tertiaryinfotech.com

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SCHEDULE

Lesson Plan — One Day, 7.5 hours

Morning

- Morning · 9:30am–12:35pm

– Topic 1: Designing a Multi AI Agents Trading Workflow (Lab 1)

Afternoon

- Afternoon · 1:05pm–5:30pm

– Topic 2: Building the Research and Decision Agents (Labs 2–5)

– Topic 3: Execution, Performance and the Feedback Loop (Labs 6–8)

- Daily timing

– 9:30am–5:30pm · 7.5 hours · 30-minute lunch

- Prerequisite

– AI Powered Quantitative Trading Fundamentals (C728) - the course that demonstrates this multi AI agents trading system; C177 teaches you to build it from scratch

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

Design the Workflow

Design a multi AI agents trading workflow: the eight agent roles, the message contract between them and a lead agent that orchestrates the flow.

2

Build the Research Agents

Vibe-code the research agents that gather Twelve Data candles, Yahoo Finance and Equibles company data, and Reddit and NewsAPI sentiment.

3

Build Decision and Execution

Vibe-code the decision and execution agents that formulate a strategy from trend, momentum, volatility, VWAP, RSI and MACD, generate long, short or null signals and place orders on an Alpaca paper account.

4

Close the Loop and Publish

Close the loop with a performance dashboard and a strategy review agent, orchestrate the full workflow and publish the project to GitHub.

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LEARNING APPROACH

How You'll Learn

1. Trainer demonstrates the concept
2. You build it yourself in the hands-on lab
3. Verify with the lab's 'Test it' step
4. Discuss and troubleshoot together
5. Recap against the learning outcomes

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CORE CONCEPTS

Course Fundamentals

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OVERVIEW

What You Will Build Today

1

Lead Agent

Orchestrates the whole flow and writes the run summary.

2

Finance Research Agent

Twelve Data 1min / 15min / 1h candles, Yahoo Finance and Equibles company data.

3

News Sentiment Agent

Reddit posts and NewsAPI headlines scored into one sentiment.

4

Trading Strategy Agent

Merges finance and sentiment; picks or mixes trend, momentum, volatility, VWAP, RSI and MACD.

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OVERVIEW

What You Will Build Today (cont.)

5

Signal Generation Agent

Turns the strategy into a long, short or null signal.

6

Stock Trader Agent

Places the order on the Alpaca paper account.

7

Performance Analysis Agent

Computes return and risk metrics for the dashboard.

8

Strategy Review Agent

Reviews the results and updates the strategy agent's guidance.

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THE BUILDING BLOCKS

The Workflow

1

1. Lead

Plans the run for each symbol.

2

2. Research

Finance Research and News Sentiment run in parallel.

3

3. Strategy

Votes from six algorithms, weighted with sentiment.

4

4. Signal

Long, short or null.

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The Workflow (cont.)

5

5. Trade

Market order on Alpaca paper.

6

6. Performance

Return, Sharpe, drawdown, win rate.

7

7. Review

New guidance for the next run - the feedback loop.

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KEY IDEA

You design the agents. The AI writes the code. The tests decide.

Every lab gives you a spec and a prompt for your AI coding assistant, then a check that proves the generated agent honours the contract.

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Your Toolkit

Build

- Python 3.11+
- Claude Code or OpenAI Codex
- Git and GitHub

Data

- Twelve Data
- Yahoo Finance (yfinance)
- Equibles
- Reddit API
- NewsAPI.org

Trade and show

- Alpaca paper account
- Streamlit dashboard
- Offline sample data

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THE LEARNING ARC

How Every Lab Progresses

1. Read the agent's spec and its message contract.
2. Prompt your AI coding assistant with the lab prompt.
3. Review the generated code against the spec.
4. Run the lab's checks offline, then with your API keys.
5. Commit to Git before the next agent.

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01

TOPIC 01

Designing a Multi AI Agents Trading Workflow

Agent roles · message contract · lead orchestration · the AI vibe coding loop

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Key Concepts — Designing a Multi AI Agents Trading Workflow

1

Agentic workflow

Several specialised agents, each with one job, its own instructions and its own data tools, coordinated by a lead agent.

2

The eight agents

Lead, Finance Research, News Sentiment, Trading Strategy, Signal Generation, Stock Trader, Performance Analysis and Strategy Review.

3

Message contract

Every agent returns one typed JSON object; the next agent reads only that object. The contract is what makes AI-written code fit together.

4

AI vibe coding loop

Spec -> prompt -> generate -> review -> test -> commit. You own the design and the checks; the coding agent writes the code.

5

Code computes, the LLM judges

Indicators, maths and order sizes are computed in code; the LLM chooses, weighs and explains. Every agent has a rule-based fallback.

6

Paper trading only

All orders go to an Alpaca PAPER account or the offline simulator. Nothing in this course is investment advice.

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WHAT YOU'LL DO

Hands-On Labs — Designing a Multi AI Agents Trading Workflow

Lab 1

- Scaffold the Project and the Agent Message Contract

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BEFORE LAB 1

Key Concepts for Lab 1

1

One role per agent

Each agent has one job, its own prompt and its own data tools - easier to prompt, test and replace.

2

Typed contract first

Write messages.py before any agent: FinanceReport, SentimentReport, Strategy, TradeSignal, OrderResult, Performance, Review.

3

LLM with a fallback

ask_json(system, payload, fallback, keys) returns the LLM's JSON, or the rule brain's output on any error - the workflow never stops.

4

Secrets stay in .env

Keys load from .env via python-dotenv; .env is git-ignored and never pasted into a prompt.

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TOPIC 01 · HANDS-ON

Scaffold the Project and the Agent Message Contract

LAB 1

Create your own project from an empty folder with your AI coding assistant: the package layout, the .env settings, the typed message contract that every agent returns, and one ask_json() helper that calls Claude or OpenAI and falls back to a rule-based brain. Load the offline sample data so every later lab runs without API keys.

You'll build

A Git repository named trading-agents with trading_agents/config.py, messages.py, llm.py, store.py, data/sample.py, sample_data/, .env.example, .gitignore and a first commit

Tools: Python 3.11+, Claude Code or OpenAI Codex, Git, python-dotenv, pandas

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LAB 1

Scaffold the Project and the Agent Message Contract

STEP 1 OF 6

1

Create the project folder, a virtual environment and Git, then open your AI coding assistant in it (Claude Code: claude, Codex: codex).

```
$ mkdir trading-agents && cd trading-agents
python -m venv .venv && source .venv/bin/activate
git init && claude
```


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LAB 1

Scaffold the Project and the Agent Message Contract

STEP 2 OF 6

2

Prompt the assistant to scaffold the project. Paste the prompt; let it create the files, then read every file it wrote.

PROMPT › Create a Python 3.11 package trading_agents/ with config.py (load .env with python-dotenv: OFFLINE, LLM_PROVIDER, ANTHROPIC_API_KEY, OPENAI_API_KEY, TWELVEDATA_API_KEY, EQUIBLES_API_KEY, NEWSAPI_KEY, REDDIT_CLIENT_ID, REDDIT_CLIENT_SECRET, … (full text in the lab README)

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LAB 1

Scaffold the Project and the Agent Message Contract

STEP 3 OF 6

3

Prompt for the message contract - the single most important file in the project.

```
PROMPT › Create trading_agents/messages.py with TypedDicts FinanceReport, SentimentReport, Strategy,
TradeSignal (signal is 'long' | 'short' | 'null', confidence 0..1), OrderResult, Performance and Review, and ALGORITHMS
= … (full text in the lab README)
```


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LAB 1

Scaffold the Project and the Agent Message Contract

STEP 4 OF 6

4

Prompt for the LLM helper with a rule-based fallback.

PROMPT › Create trading_agents/llm.py with ask_json(system, payload, fallback, keys) -> (dict, brain). Use Anthropic when LLM_PROVIDER=anthropic, OpenAI when openai, else return fallback with brain 'rules'. Tell the model to treat market data and … (full text in the lab README)

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LAB 1

Scaffold the Project and the Agent Message Contract

STEP 5 OF 6

5

Copy the course's sample data and add the offline loader and the JSON state store.

```
$ cp -r <course-repo>/labs/solution/sample_data .
PROMPT: Add trading_agents/data/sample.py (candles(symbol, interval) reads sample_data/<SYM>_<interval>.csv;
load(name, symbol) reads sample_data/<name>.json) and trading_agents/store.py … (full text in the lab README)
```


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LAB 1

Scaffold the Project and the Agent Message Contract

STEP 6 OF 6

6

Copy .env.example to .env with OFFLINE=true, run the Lab 1 check, and make your first commit.

```
$ cp .env.example .env
mkdir -p tests && cp <course-repo>/labs/solution/tests/*.py tests/
pytest -q tests/test_lab1.py
git add -A && git commit -m "Lab 1: project scaffold and agent contract"
```


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LAB 1 · VERIFY

Scaffold the Project and the Agent Message Contract

✅ Test it

tests/test_lab1.py passes: messages.py exposes the seven contracts and the six algorithms, and ask_json() returns the fallback with brain 'rules' when LLM_PROVIDER=none. git status shows .env is not tracked.

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TOPIC RECAP

Recap — Designing a Multi AI Agents Trading Workflow

1. You can now: LO1: design the eight-agent workflow, its message contract and the LLM helper every agent uses

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02

TOPIC 02

Building the Research and Decision Agents

Twelve Data · Yahoo Finance · Equibles · Reddit · NewsAPI · strategy algorithms · signals

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Key Concepts — Building the Research and Decision Agents

1

Multi-timeframe data

1-minute, 15-minute and 1-hour candles answer different questions: entry timing, intraday trend and session context.

2

Fundamentals and alternative data

Yahoo Finance company ratios and Equibles valuation, short-interest and insider data put the price action in context.

3

Sentiment as a signal

Reddit chatter and news headlines are scored from -1 to +1; news is weighted above social posts.

4

Algorithm mix

Trend, momentum, volatility breakout, VWAP, RSI and MACD each vote +1, 0 or -1; the strategy agent weights the votes and the sentiment.

5

Long, short or null

A signal fires only when the combined score clears the entry threshold; otherwise the agent stays out.

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WHAT YOU'LL DO

Hands-On Labs — Building the Research and Decision Agents

Labs 2–3

- Build the Finance Research Agent
- Build the News Sentiment Agent with Reddit and NewsAPI

Labs 4–5

- Build the Trading Strategy Agent
- Build the Signal Generation Agent

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BEFORE LAB 2

Key Concepts for Lab 2

1

Three intervals

1min for entry timing, 15min for the intraday trend the indicators use, 1h for session context.

2

Twelve Data

GET https://api.twelvedata.com/time_series with symbol, interval and an 'apikey' Authorization header.

3

Equibles

Bearer-key REST API: /stocks/{symbol}/valuation-multiples, /short-interest and /insider-transactions.

4

Indicators in code

EMA 20/50, 10-bar momentum, RSI 14, MACD 12/26/9, VWAP and Bollinger bands - computed with pandas, never by the LLM.

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TOPIC 02 · HANDS-ON

Build the Finance Research Agent

LAB 2

Direct your assistant to build three data clients - Twelve Data (1-minute, 15-minute and 1-hour candles), Yahoo Finance (company profile and ratios through yfinance) and Equibles (valuation multiples, short interest and insider transactions) - plus the indicator module, then the Finance Research Agent that combines them into one FinanceReport.

You'll build

trading_agents/data/twelvedata.py, yahoo.py, equibles.py, trading_agents/indicators.py and agents/finance_research.py

Tools: Twelve Data API, yfinance, Equibles API, pandas, Claude Code or Codex

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LAB 2

Build the Finance Research Agent

STEP 1 OF 5

1

Prompt for the Twelve Data client.

PROMPT › Create trading_agents/data/twelvedata.py with INTERVALS=('1min','15min','1h') and candles(symbol, interval) that calls https://api.twelvedata.com/time_series (outputsize 300, order asc, header Authorization: apikey <TWELVEDATA_API_KEY>) and … (full text in the lab README)

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LAB 2

Build the Finance Research Agent

STEP 2 OF 5

2

Prompt for the Yahoo Finance and Equibles clients.

PROMPT › Create data/yahoo.py company(symbol) returning shortName, sector, marketCap, trailingPE, forwardPE, profitMargins, revenueGrowth, debtToEquity, beta and 52-week range from yfinance, and data/equibles.py company(symbol) calling … (full text in the lab README)

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LAB 2

Build the Finance Research Agent

STEP 3 OF 5

3

Prompt for the indicators module.

PROMPT › Create trading_agents/indicators.py with ema, rsi, macd, vwap, bollinger, snapshot(df) returning the latest close, ema_fast(20), ema_slow(50), momentum_pct(10), rsi, macd, macd_signal, vwap, bb_low, bb_high, and votes(snapshot) returning … (full text in the lab README)

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LAB 2

Build the Finance Research Agent

STEP 4 OF 5

4

Prompt for the agent itself.

PROMPT › Create agents/finance_research.py with NAME='finance_research' and run(symbol) -> (FinanceReport, brain): fetch all three intervals, compute indicators.snapshot on the 15min candles, add Yahoo and Equibles data, build a one-line summary as … (full text in the lab README)

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LAB 2

Build the Finance Research Agent

STEP 5 OF 5

5

Review the code, run the check offline, then live with your keys, and commit.

```
$ pytest -q tests/test_lab2.py
OFFLINE=false python -c "from trading_agents.agents import finance_research as f; print(f.run('NVDA')[0]['summary'])"
git add -A && git commit -m "Lab 2: finance research agent"
```


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LAB 2 · VERIFY

Build the Finance Research Agent

✅ Test it

tests/test_lab2.py passes: the report has candles for 1min, 15min and 1h, a positive last_price, an RSI value, company and Equibles data. With keys set, the live summary names the real last price.

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BEFORE LAB 3

Key Concepts for Lab 3

1

Reddit OAuth

POST https://www.reddit.com/api/v1/access_token with grant_type=client_credentials and your app's id/secret; then call https://oauth.reddit.com with 'bearer <token>' and a descriptive User-Agent.

2

Where traders talk

Search r/stocks, r/wallstreetbets and r/investing for the symbol, newest first, past week.

3

NewsAPI

GET https://newsapi.org/v2/everything?q=<symbol> with the X-Api-Key header.

4

Blend, then refine

Lexicon scores give a deterministic base (news 60%, Reddit 40%); the LLM may refine the score but never leaves -1..1.

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TOPIC 02 · HANDS-ON

Build the News Sentiment Agent with Reddit and NewsAPI

LAB 3

Integrate the Reddit API with the OAuth application-only flow, add a NewsAPI.org client and a small finance lexicon scorer, then build the News Sentiment Agent that blends both sources into one score from -1 to +1 and lets the LLM refine it.

You'll build

trading_agents/data/reddit.py, data/newsapi.py, trading_agents/sentiment.py and agents/news_sentiment.py

Tools: Reddit API (script app), NewsAPI.org, httpx, Claude Code or Codex

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LAB 3

Build the News Sentiment Agent with Reddit and NewsAPI

STEP 1 OF 5

1

Create a Reddit 'script' app at reddit.com/prefs/apps and put the id, secret and a User-Agent in .env.

```
$ REDDIT_CLIENT_ID=<YOUR-ID>
REDDIT_CLIENT_SECRET=<YOUR-SECRET>
REDDIT_USER_AGENT=trading-agents/1.0 by <your-username>
```


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LAB 3

Build the News Sentiment Agent with Reddit and NewsAPI

STEP 2 OF 5

2

Prompt for the Reddit integration.

PROMPT › Create trading_agents/data/reddit.py: token() gets an application-only OAuth token (client_credentials) from https://www.reddit.com/api/v1/access_token; posts(symbol) searches each subreddit in REDDIT_SUBREDDITS via … (full text in the lab README)

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LAB 3

Build the News Sentiment Agent with Reddit and NewsAPI

STEP 3 OF 5

3

Prompt for the NewsAPI client and the lexicon scorer.

PROMPT › Create data/newsapi.py headlines(symbol) using https://newsapi.org/v2/everything (X-Api-Key header) and trading_agents/sentiment.py with score(text) from small positive/negative finance word lists and average(texts).

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LAB 3

Build the News Sentiment Agent with Reddit and NewsAPI

STEP 4 OF 5

4

Prompt for the agent.

PROMPT › Create agents/news_sentiment.py, NAME='news_sentiment', run(symbol) -> (SentimentReport, brain): score Reddit and news separately, combine 0.4*reddit + 0.6*news as the fallback score, ask_json() may refine score and summary, then clamp … (full text in the lab README)

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LAB 3

Build the News Sentiment Agent with Reddit and NewsAPI

STEP 5 OF 5

5

Run the check offline and live, and commit.

```
$ pytest -q tests/test_lab3.py
OFFLINE=false python -c "from trading_agents.data import reddit; print(len(reddit.posts('NVDA')))"
git add -A && git commit -m "Lab 3: news sentiment agent with Reddit"
```


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LAB 3 · VERIFY

Build the News Sentiment Agent with Reddit and NewsAPI

✅ Test it

tests/test_lab3.py passes: score is between -1 and 1 and both Reddit posts and news articles were read. With keys set, reddit.posts('NVDA') returns recent posts from the configured subreddits.

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BEFORE LAB 4

Key Concepts for Lab 4

1

Six algorithms

Trend (EMA 20 vs 50), momentum (10-bar change), volatility (Bollinger breakout), VWAP (price vs VWAP), RSI (30/70 reversal) and MACD (line vs signal).

2

Single or mix

The agent may pick one algorithm or weight several; weights are normalised to sum to 1.

3

Combined score

(1 - sentiment weight) x weighted votes + sentiment weight x sentiment score.

4

Guidance in, strategy out

DEFAULT_GUIDANCE (trend 0.4, MACD 0.35, RSI 0.25, threshold 0.3, sentiment weight 0.3) until the review agent writes guidance.json.

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TOPIC 02 · HANDS-ON

Build the Trading Strategy Agent

LAB 4

Build the agent that turns the two research reports into a strategy: the six algorithm votes come from code, the agent chooses one algorithm or a weighted mix, adds the sentiment, and reads the strategy review's guidance so the next lab's feedback loop has something to change.

You'll build

agents/trading_strategy.py with DEFAULT_GUIDANCE and a combined score from -1 to +1

Tools: indicators.py votes, ask_json(), Claude Code or Codex

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LAB 4

Build the Trading Strategy Agent

STEP 1 OF 4

1

Prompt for the agent.

```
PROMPT › Create agents/trading_strategy.py, NAME='trading_strategy', DEFAULT_GUIDANCE as described, and run(finance, sentiment, guidance) -> (Strategy, brain). Compute votes with indicators.votes() - never ask the LLM for indicator values. Let … (full text in the lab README)
```


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LAB 4

Build the Trading Strategy Agent

STEP 2 OF 4

2

Review: confirm the LLM can only choose weights, never invent votes or prices.

```
$ grep -n "votes" trading_agents/agents/trading_strategy.py
```


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LAB 4

Build the Trading Strategy Agent

STEP 3 OF 4

3

Try three mixes by hand and compare the combined score.

```
$ python -c "from trading_agents.agents import finance_research as f, news_sentiment as n, trading_strategy as t;
fi=f.run('AAPL')[0]; se=n.run('AAPL')[0]; [print(w, t.run(fi, se, {'weights': w})[0]['combined_score']) for w in ({'trend':1}, …
(full text in the lab README)
```


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LAB 4

Build the Trading Strategy Agent

STEP 4 OF 4

4

Run the check and commit.

```
$ pytest -q tests/test_lab4.py
git add -A && git commit -m "Lab 4: trading strategy agent"
```


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LAB 4 · VERIFY

Build the Trading Strategy Agent

✅ Test it

tests/test_lab4.py passes: votes cover all six algorithms and the strategy's weights sum to 1. Changing the weights changes combined_score.

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BEFORE LAB 5

Key Concepts for Lab 5

1

Threshold rule

long if score >= threshold, short if score <= -threshold, otherwise null.

2

Confidence

The size of the combined score, clamped to 0..1.

3

Guard the output

Any answer other than long, short or null is replaced by the rule signal.

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TOPIC 02 · HANDS-ON

Build the Signal Generation Agent

LAB 5

Build the small but critical agent that converts the combined score into exactly one signal with a confidence, using the entry threshold from the guidance, and guard against any invalid LLM answer.

You'll build

agents/signal_generation.py with rule_signal() and run()

Tools: ask_json(), Claude Code or Codex

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LAB 5

Build the Signal Generation Agent

STEP 1 OF 3

1

Prompt for the agent.

```
PROMPT › Create agents/signal_generation.py, NAME='signal_generation', rule_signal(score, threshold) and
run(strategy) -> (TradeSignal, brain) as specified; validate the LLM's signal and clamp confidence to 0..1.
```


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LAB 5

Build the Signal Generation Agent

STEP 2 OF 3

2

Run the check and print the signals for three symbols.

```
$ pytest -q tests/test_lab5.py
python -c "from trading_agents.agents import signal_generation as s; print(s.rule_signal(0.5,0.3),
s.rule_signal(-0.5,0.3), s.rule_signal(0.1,0.3))"
```


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LAB 5

Build the Signal Generation Agent

STEP 3 OF 3

3

Commit.

```
$ git add -A && git commit -m "Lab 5: signal generation agent"
```


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LAB 5 · VERIFY

Build the Signal Generation Agent

✅ Test it

tests/test_lab5.py passes and the three sample scores print long, short and null.

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TOPIC RECAP

Recap — Building the Research and Decision Agents

1. You can now: LO2: vibe-code an agent that gathers Twelve Data candles, Yahoo Finance and Equibles company data

2. You can now: LO2: vibe-code the Reddit API integration and an agent that scores Reddit and news sentiment

3. You can now: LO3: vibe-code an agent that merges finance and sentiment into a weighted mix of trading algorithms

4. You can now: LO3: vibe-code an agent that turns the strategy into a long, short or null signal

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03

TOPIC 03

Execution, Performance and the Feedback Loop

Alpaca paper orders · performance metrics · strategy review · lead orchestration · GitHub

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Key Concepts — Execution, Performance and the Feedback Loop

1

Automated paper execution

The stock trader agent turns a signal into a market order on Alpaca paper: long buys, short sells short, null does nothing.

2

Performance metrics

Total return, Sharpe ratio, maximum drawdown and win rate measure whether the agents add value.

3

Feedback loop

The strategy review agent reads the performance and rewrites the guidance the strategy agent uses next run.

4

Orchestration

The lead agent plans the run, runs the two research agents in parallel and the rest in sequence, and writes the summary.

5

Publish safely

Push to GitHub with .env ignored, keys in environment variables only, a README and passing tests.

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WHAT YOU'LL DO

Hands-On Labs — Execution, Performance and the Feedback Loop

Lab 6

- Build the Stock Trader Agent on Alpaca Paper

Lab 7

- Build the Performance Analysis Agent and the Dashboard

Lab 8

- Close the Loop: Strategy Review, Lead Agent and GitHub

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BEFORE LAB 6

Key Concepts for Lab 6

1

Paper only

TradingClient(key, secret, paper=True) - the code never touches a live account.

2

Long, short, flip

long -> buy; short -> sell short; an opposite position is closed first; already in the same direction -> skip.

3

Position sizing

Shares = equity x POSITION_PCT / price, rounded down.

4

Same interface offline

PaperSim keeps cash, positions and fills in state/ so the agent behaves identically without keys.

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TOPIC 03 · HANDS-ON

Build the Stock Trader Agent on Alpaca Paper

LAB 6

Build the broker layer - Alpaca paper trading through alpaca-py, with an offline simulated paper broker of the same interface - and the Stock Trader Agent that sizes the position and buys, sells short, flips or stays out according to the signal.

You'll build

trading_agents/broker.py (AlpacaPaper, PaperSim, get_broker) and agents/stock_trader.py

Tools: Alpaca paper trading API (alpaca-py), Claude Code or Codex

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LAB 6

Build the Stock Trader Agent on Alpaca Paper

STEP 1 OF 5

1

Create a free Alpaca paper account, copy the paper keys into .env.

```
$ ALPACA_API_KEY=<PAPER-KEY>
ALPACA_SECRET_KEY=<PAPER-SECRET>
```


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LAB 6

Build the Stock Trader Agent on Alpaca Paper

STEP 2 OF 5

2

Prompt for the broker layer.

PROMPT › Create trading_agents/broker.py with PaperSim (cash 100000, positions with avg price, mark(), position(), positions(), equity(), submit(symbol, side, qty, price) returning status and realized P&L) and AlpacaPaper using alpaca-py … (full text in the lab README)

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LAB 6

Build the Stock Trader Agent on Alpaca Paper

STEP 3 OF 5

3

Prompt for the agent.

PROMPT › Create agents/stock_trader.py, NAME='stock_trader', run(signal, price, broker=None) -> (OrderResult, 'code'): null -> no order; same direction already held -> skip; otherwise close the opposite position, then buy (long) or sell short … (full text in the lab README)

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LAB 6

Build the Stock Trader Agent on Alpaca Paper

STEP 4 OF 5

4

Run the check offline, then place one live paper order and find it in the Alpaca dashboard.

```
$ pytest -q tests/test_lab6.py
OFFLINE=false python -c "from trading_agents.agents import stock_trader as t;
print(t.run({'symbol':'SPY','signal':'long'}, 580.0)[0])"
```


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LAB 6

Build the Stock Trader Agent on Alpaca Paper

STEP 5 OF 5

5

Commit.

```
$ git add -A && git commit -m "Lab 6: stock trader agent on Alpaca paper"
```


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LAB 6 · VERIFY

Build the Stock Trader Agent on Alpaca Paper

✅ Test it

tests/test_lab6.py passes: long opens a long position, short closes it and opens a short, null places no order. The live paper order appears under Orders in the Alpaca paper dashboard.

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BEFORE LAB 7

Key Concepts for Lab 7

1

Total return

Latest equity / starting equity - 1.

2

Sharpe ratio

Mean of per-run returns / their standard deviation x sqrt(252).

3

Maximum drawdown

The worst fall from a running equity peak.

4

Win rate

Share of closing trades with positive realised P&L.

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TOPIC 03 · HANDS-ON

Build the Performance Analysis Agent and the Dashboard

LAB 7

Build the agent that records the equity after every run and computes total return, Sharpe ratio, maximum drawdown, win rate and trade count, then a Streamlit dashboard that runs the workflow and shows the metrics, the equity curve, the signals and the orders.

You'll build

agents/performance_analysis.py (metrics() and run()) and dashboard.py

Tools: Streamlit, pandas, Claude Code or Codex

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LAB 7

Build the Performance Analysis Agent and the Dashboard

STEP 1 OF 4

1

Prompt for the agent.

PROMPT › Create agents/performance_analysis.py, NAME='performance_analysis', metrics(curve, trades) returning total_return_pct, sharpe, max_drawdown_pct, win_rate_pct and trades, and run(broker=None) that appends the equity to equity.jsonl, computes … (full text in the lab README)

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LAB 7

Build the Performance Analysis Agent and the Dashboard

STEP 2 OF 4

2

Prompt for the dashboard.

PROMPT › Create dashboard.py in Streamlit: a symbols box and a 'Run the workflow' button, a five-metric row (equity, return, Sharpe, max drawdown, win rate) from performance.json, a line chart of equity.jsonl, and the strategy guidance from … (full text in the lab README)

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LAB 7

Build the Performance Analysis Agent and the Dashboard

STEP 3 OF 4

3

Run the check and open the dashboard.

```
$ pytest -q tests/test_lab7.py
streamlit run dashboard.py
```


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LAB 7

Build the Performance Analysis Agent and the Dashboard

STEP 4 OF 4

4

Commit.

```
$ git add -A && git commit -m "Lab 7: performance analysis agent and dashboard"
```


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LAB 7 · VERIFY

Build the Performance Analysis Agent and the Dashboard

✅ Test it

tests/test_lab7.py passes: a rising equity curve gives a positive return and zero drawdown; a fall gives a negative drawdown. The dashboard shows the five metrics after a run.

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BEFORE LAB 8

Key Concepts for Lab 8

1

Review rules

With two or more trades and a negative return, raise the entry threshold by 0.05 and adjust; otherwise keep. The LLM may re-weight the algorithms.

2

guidance.json

The review writes it; the strategy agent reads it on the next run - that is the feedback loop.

3

Lead orchestration

Plan -> [finance research || news sentiment] -> strategy -> signal -> trader per symbol -> performance -> review -> summary.

4

Trace everything

Every agent call is one trace entry: agent, brain (LLM or rules) and output.

5

Publish

A README with the flow diagram and set-up, .env.example but never .env, tests passing, then git push.

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TOPIC 03 · HANDS-ON

Close the Loop: Strategy Review, Lead Agent and GitHub

LAB 8

Build the Strategy Review Agent that reads the performance and rewrites the strategy guidance, and the Lead Agent that orchestrates the whole workflow - the two research agents in parallel, then strategy, signal, trade, performance and review - and summarises the run. Run it twice to watch the loop change the strategy, then push your project to your own GitHub repository.

You'll build

agents/strategy_review.py, agents/lead.py, run.py, a README and a public GitHub repository

Tools: ThreadPoolExecutor, Git, GitHub, Claude Code or Codex

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LAB 8

Close the Loop: Strategy Review, Lead Agent and GitHub

STEP 1 OF 5

1

Prompt for the review agent.

PROMPT › Create agents/strategy_review.py, NAME='strategy_review', run(performance, strategies) -> (Review, brain): start from DEFAULT_GUIDANCE merged with guidance.json; if trades >= 2 and total return < 0 set verdict 'adjust' and raise … (full text in the lab README)

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LAB 8

Close the Loop: Strategy Review, Lead Agent and GitHub

STEP 2 OF 5

2

Prompt for the lead agent and the CLI.

PROMPT › Create agents/lead.py, NAME='lead', run(symbols) that records a plan trace entry, then for each symbol runs finance_research and news_sentiment in parallel with ThreadPoolExecutor, then trading_strategy (with guidance.json), … (full text in the lab README)

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LAB 8

Close the Loop: Strategy Review, Lead Agent and GitHub

STEP 3 OF 5

3

Run the whole workflow twice and compare: the second run reads the review's guidance.

```
$ python run.py NVDA AAPL SPY
python run.py NVDA AAPL SPY
cat state/guidance.json
```


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LAB 8

Close the Loop: Strategy Review, Lead Agent and GitHub

STEP 4 OF 5

4

Run every check.

```
$ pytest -q
```


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LAB 8

Close the Loop: Strategy Review, Lead Agent and GitHub

STEP 5 OF 5

5

Ask the assistant for a README, create an empty GitHub repository and push.

```
PROMPT › Write README.md with the eight agents, a mermaid flow diagram, set-up (venv, .env from .env.example,
OFFLINE=true), how to run run.py, the dashboard and the tests.
git add -A && git commit -m "Lab 8: review loop, lead agent, README"
git … (full text in the lab README)
```


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LAB 8 · VERIFY

Close the Loop: Strategy Review, Lead Agent and GitHub

✅ Test it

pytest -q passes every check; the trace lists lead, finance_research, news_sentiment, trading_strategy, signal_generation, stock_trader, performance_analysis, strategy_review and lead; the second run's strategy uses the threshold in guidance.json; your repository is on GitHub without a .env file.

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TOPIC RECAP

Recap — Execution, Performance and the Feedback Loop

1. You can now: LO3: vibe-code an agent that places long and short market orders on an Alpaca paper account

2. You can now: LO4: vibe-code an agent that computes return and risk metrics and a dashboard that shows them

3. You can now: LO4: vibe-code the strategy review feedback loop and the lead agent, then publish the project to GitHub

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

Design the Workflow

Design a multi AI agents trading workflow: the eight agent roles, the message contract between them and a lead agent that orchestrates the flow.

2

Build the Research Agents

Vibe-code the research agents that gather Twelve Data candles, Yahoo Finance and Equibles company data, and Reddit and NewsAPI sentiment.

3

Build Decision and Execution

Vibe-code the decision and execution agents that formulate a strategy from trend, momentum, volatility, VWAP, RSI and MACD, generate long, short or null signals and place orders on an Alpaca paper account.

4

Close the Loop and Publish

Close the loop with a performance dashboard and a strategy review agent, orchestrate the full workflow and publish the project to GitHub.

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NEXT STEPS

Continuing Your Journey

1. Compare your agents with the multi AI agents system demonstrated in the prerequisite course C728.

2. Add a risk-management agent and backtest the workflow before any live use.

3. Schedule the lead agent to run every 15 minutes during the US session on paper.

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WRAP-UP

Keep Practising After the Course

1. Re-run the labs on your own machine — repetition is what makes the skills stick.
2. Apply the techniques to a real process or project in your own organisation.
3. Your course material stays available at https://lms-tms.tertiaryinfotech.com/.
4. Explore the follow-on courses at www.tertiarycourses.com.sg.

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THANK YOU FOR ATTENDING

Thank You!

You built a self-improving multi AI agents trading workflow from scratch and published it on GitHub.

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